# NATIONAL COLLEGE OF IRELAND

MSc in Business Analytics for Decision Makers (MSCBADM)

## Mock Examination, April 2026

Data Visualisation (9DATAV)

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| **Module** | 9DATAV Data Visualisation |
| **Programme** | MSc in Business Analytics for Decision Makers (MSCBADM) |
| **Lecturer / Examiner** | Victor del Rosal |
| **Duration** | 2 hours |
| **Total marks** | 100 marks (Section A: 30; Section B: 70) |
| **Exam weighting** | 40% of overall module grade |

**Instructions.** This paper has two sections. **Section A** is compulsory (30 marks): answer all four questions. Questions A1 and A4 offer a choice between two paths; answer one path only for each. **Section B** (70 marks): answer two of five questions (35 marks each). Where a question offers an Applied Path inviting reflection on the DataVis Challenge or other CA work, students who did not complete the challenge (or who prefer not to reference it) may answer the Theory Path instead. This applies to A1, A4, and B1.

Credit will be given for reference to named visualisation principles, relevant theories of perception, and specific examples drawn from business analytics contexts or your DataVis Challenge work. Be specific: name the principle, theory, or design choice. Generic answers will receive limited credit.

**Note.** This mock examination mirrors the format and structure of the terminal examination. It is calibrated to be slightly more demanding in its scenarios so that students who handle it well can sit the May terminal with confidence.

**Suggested time allocation:** Section A approximately 30 minutes (7-8 minutes per question); Section B approximately 45 minutes per question.

## SECTION A: SHORT-FORM QUESTIONS (30 MARKS)

**Answer ALL FOUR questions in this section.**

### Question A1 (8 marks) (LO1)

*Answer EITHER (i) OR (ii).*

#### (i) Theory Path

A senior analyst opens an unfamiliar 18-month customer transaction dataset and rapidly produces a dozen scatter plots, histograms, and small multiples to look for unusual patterns. Two weeks later, the same analyst publishes one polished chart for the executive board.

(a) Name the two modes of visualisation the analyst is moving between, and define each precisely. (4 marks)

(b) Identify two design choices that should change between the first activity and the second, and explain why each change is justified by the shift in audience and purpose. (4 marks)

#### (ii) Applied Path

Reflect on your DataVis Challenge work.

(a) Identify one chart or view in your submission that was primarily exploratory (built for your team's understanding) and one that was primarily explanatory (built for the audience). Explain the design differences between them. (4 marks)

(b) With hindsight, name one explanatory chart in your submission that still carried exploratory residue (too much data, too many encodings, no clear message) and explain how you would simplify it. (4 marks)

### Question A2 (7 marks) (LO2)

A dashboard places a single red data point among 200 grey points. A user notices the red point in under a second, without searching.

(a) Name the perceptual mechanism that explains this and identify the visual attribute being exploited. Cite the relevant author. (3 marks)

(b) Name two further visual attributes that operate in this same way, and for each, give one business-analytics scenario where it would be the appropriate choice for drawing attention to a critical data point. (4 marks)

### Question A3 (8 marks) (LO1, LO5)

**Practical Interpretation.** A monthly sales report contains a 3D pie chart of revenue share across six product categories. The chart sits on a textured photographic background, uses a different bright hue for each slice, includes a drop shadow on each segment, and shows percentage labels both inside each slice and again in a legend to the right.

(a) Using Tufte's data-ink ratio, identify three specific elements that should be removed or simplified, and briefly justify each removal. (4 marks)

(b) Pie charts are widely criticised in the data visualisation literature for tasks of comparison. Drawing on Cleveland and McGill (1984), explain why a pie chart is a poor choice for comparing six product categories, and recommend a more accurate chart type with justification. (4 marks)

### Question A4 (7 marks) (LO2, LO5)

*Answer EITHER (i) OR (ii).*

#### (i) Theory Path

A public agency uses a rainbow palette (red-orange-yellow-green-blue-indigo-violet) for a choropleth map showing carbon emissions intensity across 50 manufacturing plants on a continuous scale.

(a) Identify two distinct perceptual problems with this palette choice for continuous data, and name the relevant authority for each. (4 marks)

(b) Recommend a more appropriate palette family with justification, and identify one accessibility consideration the redesign should also address. (3 marks)

#### (ii) AI Praxis Path

During the module, students used AI tools to generate, critique, or refine visualisations.

(a) Describe one situation (from your DataVis Challenge or class exercises) where an AI tool produced a chart or recommendation that was technically correct but visually or perceptually flawed. What was the flaw? (4 marks)

(b) What does this episode tell you about the limits of AI-assisted visualisation, and what minimum human review steps would you build into a team workflow that uses such tools? (3 marks)

## SECTION B: LONG-FORM QUESTIONS (70 MARKS)

**Answer TWO questions from a choice of FIVE. Each question is worth 35 marks.**

### Question B1 (35 marks) (LO1, LO5)

**Reflecting on the DataVis Challenge: What Your Design Did Not Show.** During the DataVis Challenge you scoped a question, sourced or selected a dataset, designed a visual artefact (newspaper, dashboard, prototype, or composite), and presented it to peers and panel.

*Answer EITHER (i) Applied Path OR (ii) Theory Path.*

#### (i) Applied Path

(a) Describe the central insight your design was built to communicate. Was it something the audience likely already suspected, or did the visualisation surface something genuinely non-obvious? Be honest. (10 marks)

(b) Identify one design decision in your submission that you would now reverse, and explain why. Reference at least one named principle or perceptual theory in your justification. (15 marks)

(c) Reflect on the gap between what your dataset could support and what your design actually claimed. Did your visualisation imply a level of certainty, completeness, or causal explanation that the underlying data could not bear? Connect this to the broader ethical challenge of visualisation as persuasion. (10 marks)

#### (ii) Theory Path

(a) “Every visualisation is an editorial act.” Drawing on Kirk (2019) and Knaflic (2015), critically discuss this claim. (10 marks)

(b) A consultancy publishes a chart that omits two quarters of negative growth from a ten-quarter dataset, titled “Revenue Growth: A Consistent Upward Trend.” Evaluate this as a violation of graphical integrity, citing relevant theory, and explain why selective omission is harder for an audience to detect than overt distortion of axis or scale. (15 marks)

(c) Discuss the ethical responsibilities of the visualisation designer when communicating to non-technical audiences who will not (and cannot reasonably be expected to) audit the underlying data. (10 marks)

### Question B2 (35 marks) (LO2, LO5)

> **Scenario: Iverna Health Trust.** Iverna is a publicly funded regional health service in the west of Ireland. Its analytics team has built a single Power BI screen for the clinical operations centre. The screen contains 16 panels: live A&E waiting times by triage category, ambulance arrivals, theatre utilisation, ward occupancy, staffing rosters, weather alerts, infection control flags, equipment maintenance status, transfer requests, discharge readiness, GP referral inflow, complaint counts, lab turnaround, radiology backlog, pharmacy stock, and a rolling news ticker. The most clinically critical panel (sepsis early warning) is in the bottom-right corner in a pale teal that matches three other non-urgent panels.
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> In the last quarter, two sepsis alerts were missed for over 40 minutes during busy periods. The clinical lead has asked for a redesign.

(a) Using named theories of perception and cognition, evaluate why this dashboard is failing clinical staff. Address at least three distinct mechanisms (for example: cognitive load and working memory, pre-attentive processing, visual hierarchy, figure-ground, or Few's principle of appropriate emphasis). (15 marks)

(b) Propose a redesigned dashboard. Address: layout and visual hierarchy with a justified primary zone; the use of pre-attentive channels for the sepsis alert; how progressive disclosure (Shneiderman) would manage the secondary panels; how you would involve clinical staff in the design process; and how the redesign would be validated before deployment. (20 marks)

### Question B3 (35 marks) (LO1, LO2, LO5)

> **Scenario: Atlas FoodCo.** Atlas FoodCo is an Irish ready-meals manufacturer selling through national supermarket chains. Sales have grown 15% year on year, but margin has fallen. The CFO has commissioned a Power BI reporting suite from an external agency to “find out where the margin is going.” The agency has data from the ERP (production cost, batch yields), the logistics partner (delivery routes, fuel, spoilage), and the retailers (weekly sales by SKU, promotional activity, returns). Three audiences have been identified: the CEO (strategic monthly view), the operations director (weekly bottleneck triage), and category managers (daily SKU-level trading decisions).
>
> The agency proposes “one master dashboard for everyone, with filters.”

(a) Critically evaluate the “one dashboard for everyone” proposal. Identify the specific cognitive and design problems it creates for each of the three audiences, and contrast with a layered or audience-segmented architecture. Reference named principles. (15 marks)

(b) For each of the three audiences, recommend chart types, the appropriate level of interactivity, and the dominant visualisation principle that should govern the design. Justify your recommendations. (20 marks)

### Question B4 (35 marks) (LO1, LO2, LO5)

> **Scenario: Meridian Wealth.** Meridian is a Dublin asset manager. A junior analyst uses an AI-powered analytics tool to generate charts for an investor presentation. One chart shows quarterly fund returns on a Y-axis starting at 4.6%, making a 0.4-percentage-point spread between funds appear as a near-vertical staircase. A second chart uses a diverging red-to-green palette for absolute returns, where the colour midpoint coincidentally falls at the median fund and visually divides “winners” from “losers” without any analytical justification. When challenged, the analyst replies: “The numbers are correct. The labels are visible. The AI generated the chart based on the data.”

(a) Critically evaluate the analyst's defence. Address: the distinction between data accuracy and graphical integrity (Tufte); why pre-attentive perception of shape and colour overrides the careful reading of axis labels (Ware; Cleveland and McGill); the inappropriate use of a diverging palette for non-diverging data; the ethical duties owed to investors; and the responsibility a human reviewer carries when an AI tool generates the artefact. (20 marks)

(b) Propose a quality assurance framework for visualisations destined for external audiences. Include at least four evaluation criteria, each grounded in a named design principle or theory, and explain how each criterion would be applied in practice. (15 marks)

### Question B5 (35 marks) (LO1, LO2, LO5)

> **Scenario: HarbourCity Open Data.** HarbourCity Council is launching a public-facing real-time dashboard for traffic congestion across a metropolitan area. Three audiences have been identified: commuters checking conditions before leaving home, journalists covering infrastructure performance, and city planners evaluating long-term transport investment. A community group has separately raised concerns that the dashboard's headline metric (“congestion index”) is computed in a way that systematically underrepresents bus-corridor delays in lower-income districts, because the underlying sensor network is sparser there. The council's communications team wants the launch to be “clean and simple, with no clutter.”

(a) Critically evaluate the design challenge of serving three distinct audiences with one public-facing system. Recommend an audience-appropriate architecture, drawing on named principles (for example, Shneiderman's mantra, progressive disclosure, audience-first design). (15 marks)

(b) Evaluate the community group's concern as a problem of visualisation ethics, not just data quality. Address: the relationship between what the underlying data can support and what the visualisation is allowed to imply; how the “clean and simple” instruction can itself become an ethical failure when it suppresses caveats and uncertainty; and what specific design and disclosure features the dashboard should include to address both the technical limitation and the public trust dimension. (20 marks)

**END OF EXAMINATION PAPER · TOTAL 100 MARKS**